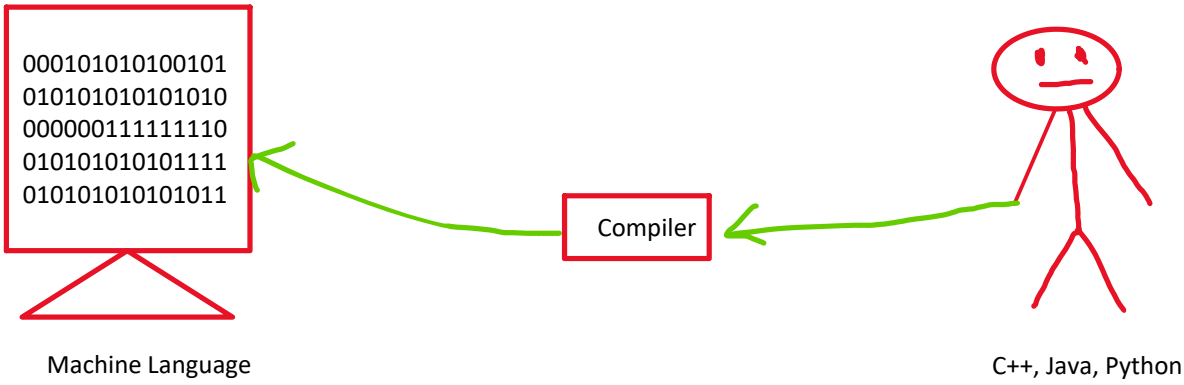
Day - 2 (FlowChart & Pesudo Code)

16 October 2024 22:28



- **Steps To Write Algorithms:**
  - Understand
  - Given Value
  - Approach
  - Code
  - Error/Debug
  - Other Solutions

- **Instruction Gave To Machine:**
  - Take Example Of Coffee Machine:

    1. Add water
    Use the markers on the machine to measure the correct amount of fresh, cold water to add to the reservoir.

    2. Insert filter
    Make sure the filter is the right size and shape for your machine.

    3. Add coffee
    Use 2 tablespoons (10 grams) of medium ground coffee per 6 fluid ounce cup.

    4. Start brewing
    The coffee is ready when it stops dripping into the server. Interrupting the brewing process can affect the flavor.

    5. Pour and enjoy
    For best taste, drink the coffee within 20 minutes or keep it warm in an insulated container.

★ FlowChart & Pesudo Code:

Good, logical programming is developed through good pre-code planning and organization. This is assisted by the use of pseudocode and program flowcharts.

★ FlowCharts:

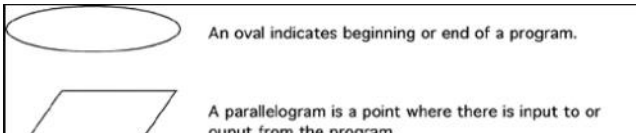
Flowcharts are written with program flow from the top of a page to the bottom. Each command is placed in a box of the appropriate shape, and arrows are used to direct program flow.

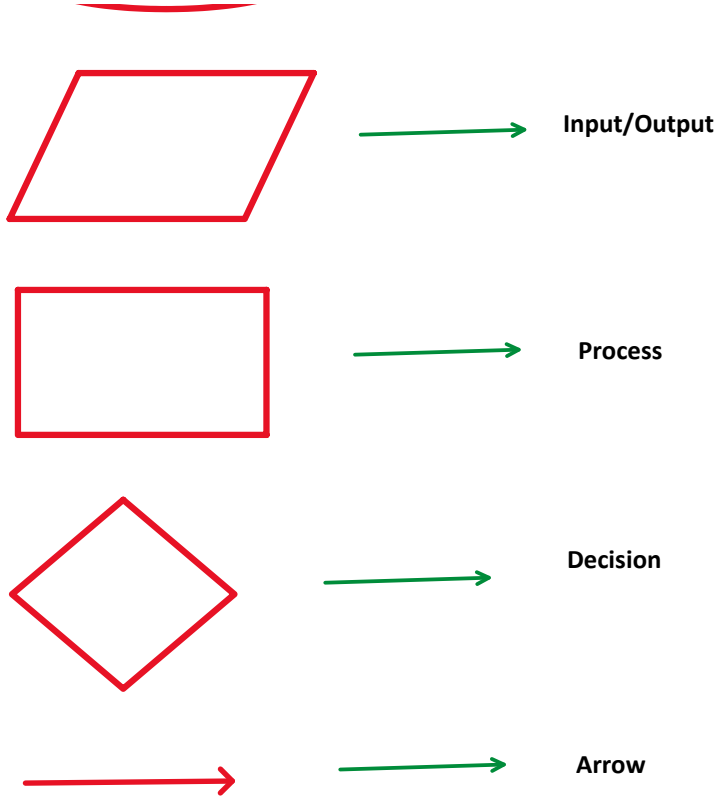
★ Pesudo Code:

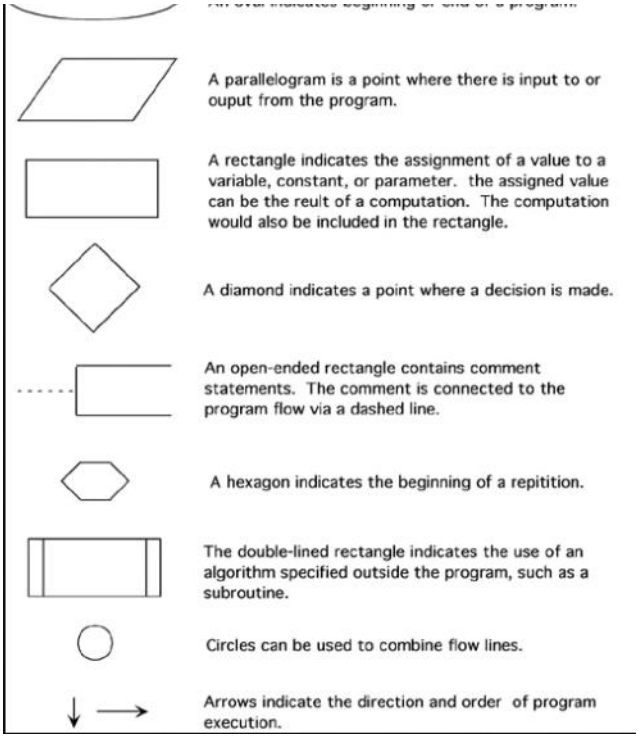
Pseudocode is a method of describing computer algorithms using a combination of natural language and programming language. It is essentially an intermittent step towards the development of the actual code. It allows the programmer to formulate their thoughts on the organization and sequence of a computer algorithm without the need for actually following the exact coding syntax.

★ Symbols In FlowChart:

Terminal

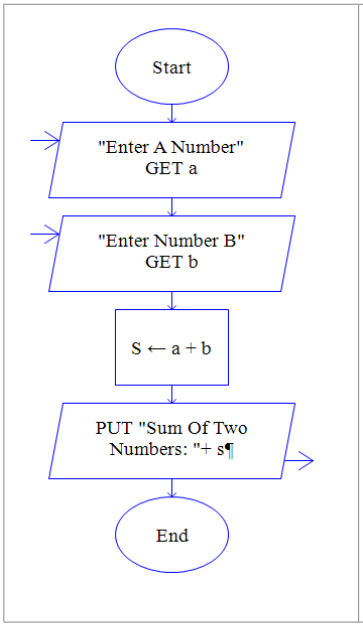






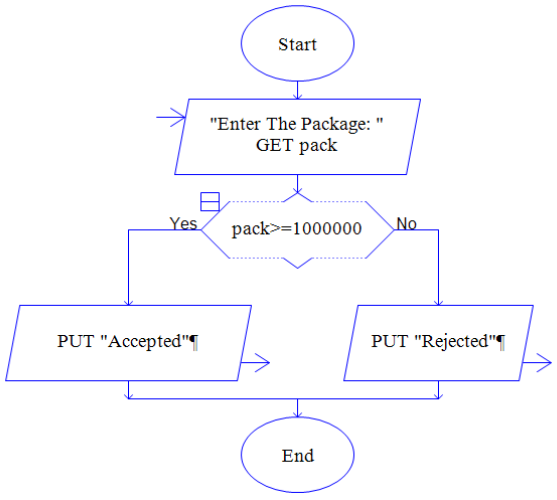
★ Draw Flow Charts:

Sum Of Two Numbers:



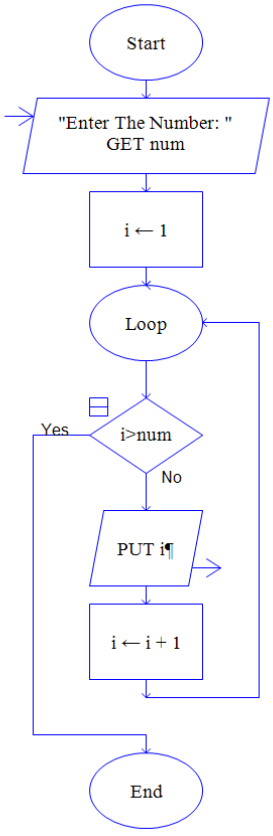
- **First**, start with terminal (Start).
- **Second & Third**, It is (Input/Output) box.
- **Fourth**, Process the input and perform arithmetic operations.
- **Fifth**, Display output of the solution.
- **Sixth**, End the problem.

Package Accept Or Reject: (Decision Making)



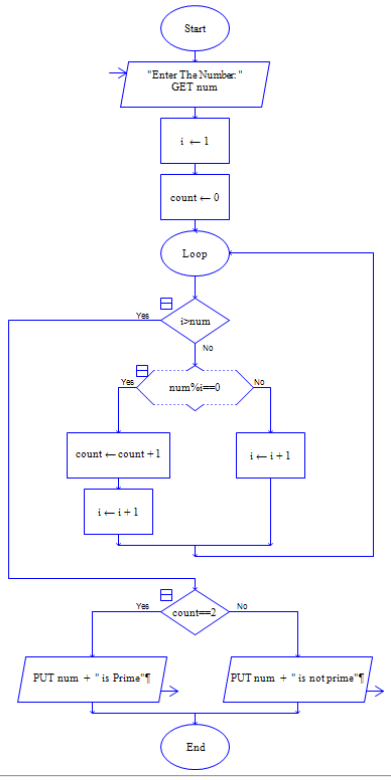
1. Start
2. Take Input From User
3. **It is decision making in which,** If Package Is Greater The 10 Lakhs Then it go to **yes** side and Accept, Otherwise go to **No** side and Reject.
4. Last Is End The Program.

Print N Numbers: (Loop)



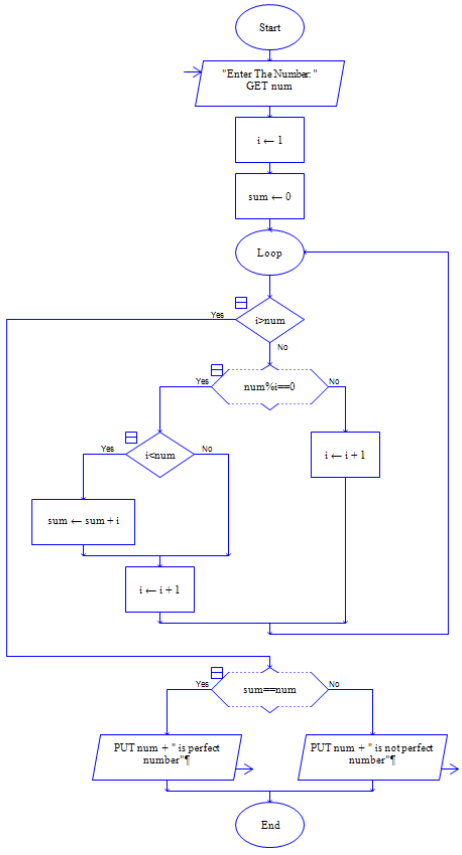
1. Start the program.
2. Take input from the user.
3. Set I is 1 (I = 1)
4. Start the loop
5. Set condition inside loop (if I is smaller than num value then it go to **No** side and display I value.
6. After display, plus 1 in I and go to top of loop again.
7. When I is greater than num then it go to **Yes** side.
8. End The Program.

★ Prime Number:



1. Start The Program.
2. Take input from user.
3. Set I = 1.
4. Set Count = 1
5. Add loop I is smaller then num then go to **No** side and inside loop use decision making if(num% i==0), then go to **Yes** side and increment count .
6. Increment i+1
7. Go to top of loop again and repeat process until it can be **False** and go to **Yes** side of loop.
8. Here also a one more **Decision Making** Statement in which, if count == 2 then it is prime. Otherwise, it in not prime number.
9. End The Program.

★ Perfect Number:



1. Start The Program.
2. Take input from user.
3. Set I =1
4. Set sum =0
5. Start loop, first I is smaller then num then go to **No** side and inside loop add **Decision Making Statement**, in this we add condition num%i==0 then go to **Yes** side and at this also add **Decision making statement** in which add I to sum if num is divide by i.
6. Increment i++
7. Repeat Until **i>num** condition is **False**.
8. At last, add **Decision Making Statement** in which condition is **(sum == num)** if yes then it is **perfect** . Otherwise, it is **not perfect**.